

SHARED FE Data Model for Architecture Center

Databricks Lakehouse Business Data Models

A library of forty production-ready Silver-layer business data models, one per industry, that deploy directly into Unity Catalog as the analytical foundation of a Databricks lakehouse.

Each model is complete, governed, and internally consistent on day one. Every domain, table, column, foreign key, classification tag, and metric view is already defined. Generic industry templates average every business in a sector, leaving customers months of trimming work. A Lakehouse Business Data Model is shaped like one organization in its industry, with the terminology and divisions that the organization actually uses.

Every model ships in two scopes, MVM (Minimum Viable Model) and ECM (Expanded Coverage Model), supports three Unity Catalog layouts (cataloging styles), and includes the same complete artifact bundle.

Where it sits



A

Lakehouse Business Data Model is the **Silver layer**. Bronze handles raw ingestion (Lakeflow, Auto Loader). Silver is the conformed, normalized analytical model that every analyst, BI tool, and ML workload reads from. Gold is derived insight (KPI tables, feature tables, aggregates) computed on top of Silver.

What ships with each model

Each model is published as a complete bundle.

``model.json`` is the logical model, capturing every domain, subdomain, product, attribute, foreign key, classification tag, and metric view definition. It is the unit of currency. Check it into git, diff across versions, share across environments.

A Unity Catalog deployment of schemas, Delta tables, primary-key constraints, foreign-key constraints (informational), and classification tags. The exact catalog and schema names depend on the cataloging style chosen at install time.

Metric views are reusable KPI definitions installed on top of the physical tables, ready for AI/BI dashboards and AI/BI Genie.

An RDFS knowledge graph (`ontology/`) expresses the same model as a semantic graph for semantic-tooling integration and AI-agent grounding.

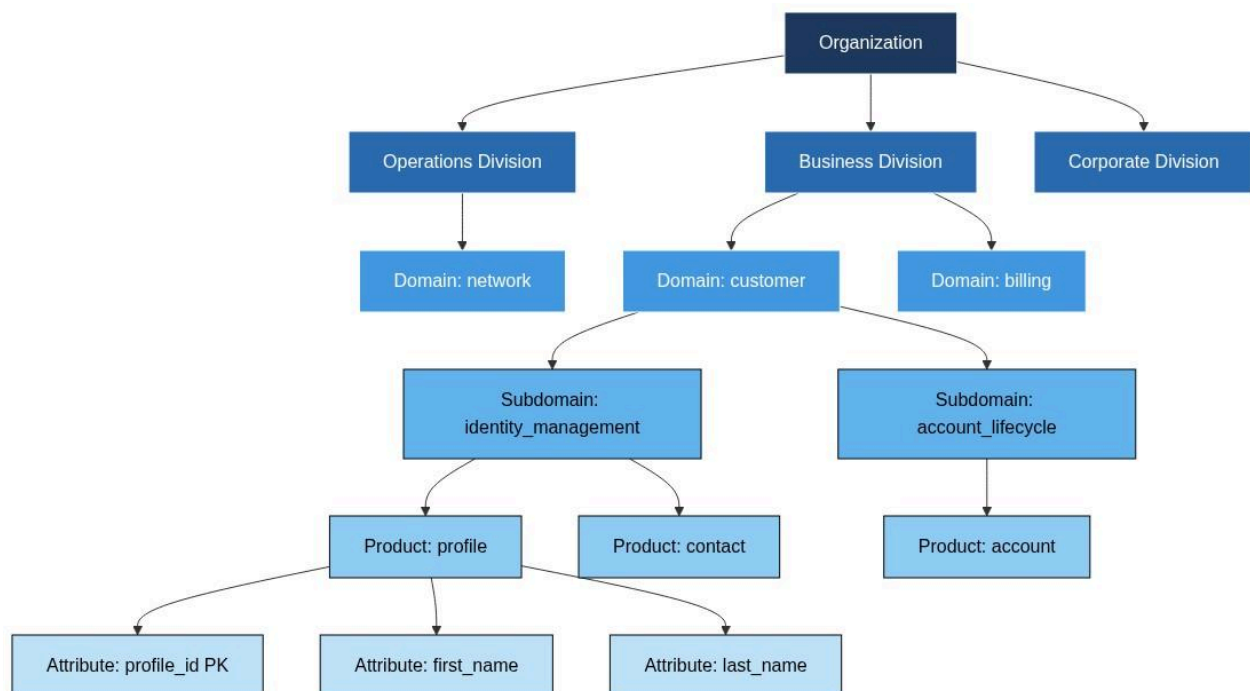
A DBML diagram (`diagram/`) for visual exploration in dbdiagram.io or any DBML-compatible viewer.

SQL DDL (`schemas/`) for the entire deployment, organized by schema.

Excel and Markdown documentation generated alongside the model.

Optional synthetic sample data. Pool-based rows respect every foreign key, regex constraint, and classification tag.

The four-level hierarchy

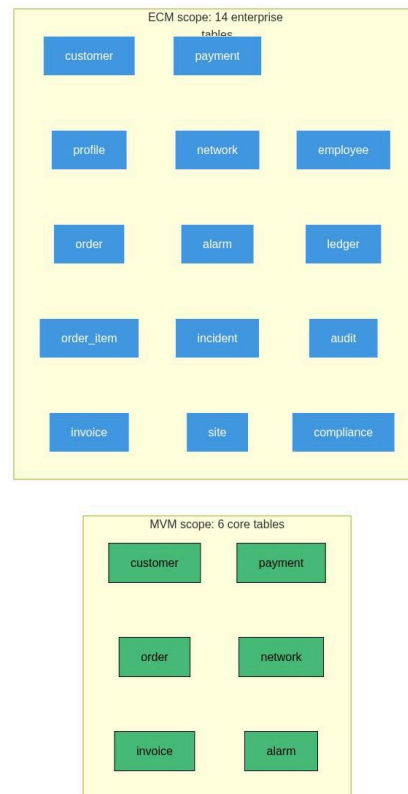


Every model follows the same hierarchy. The **organization** is the whole business. It contains three divisions: Operations (what the business physically does), Business (who it serves and how it earns), and Corporate (the supporting back office). Each division contains one or more **domains** (one-word, lowercase, singular bounded contexts). Each domain contains two or more **subdomains** (two-word semantic groupings). Each subdomain contains **products** (the Delta tables). Each product contains **attributes** (the columns), with data types and classification tags assigned up front.

Operations and Business combined always hold at least 80% of the domains; Corporate is capped at 20%. This ratio keeps every model centered on what actually runs the business rather than on its administrative back office.

Every model is also a **Directed Acyclic Graph** by construction. Foreign keys always point child-to-parent (order.customer_id, never customer.latest_order_id); cycles are broken before publication; every domain connects to at least one other through foreign keys, so cross-domain analytics is always possible.

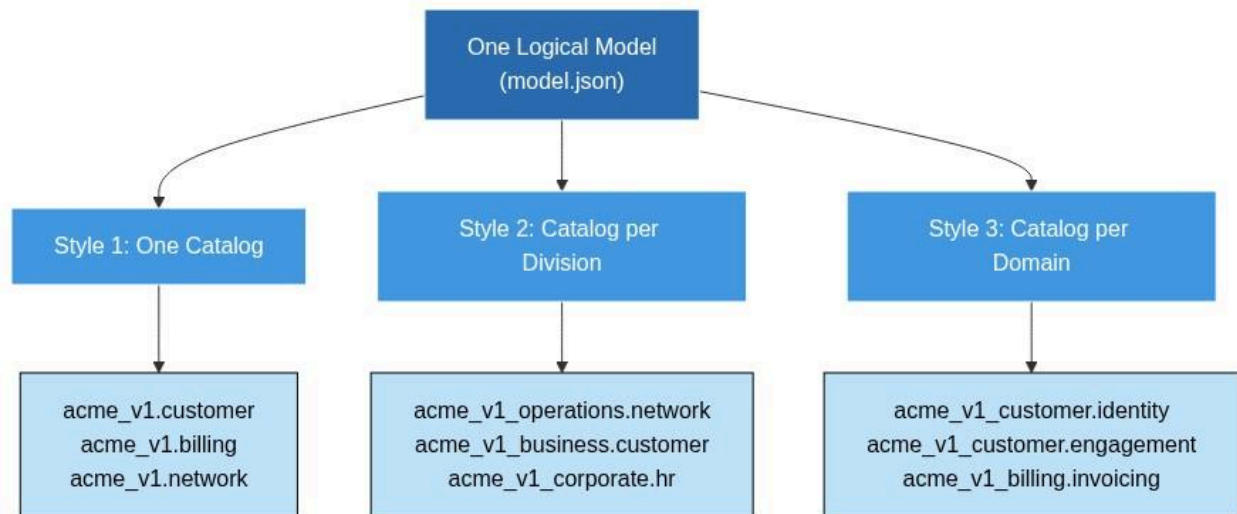
Two scopes: MVM and ECM



MVM (Minimum Viable Model) carries 30 to 50 percent of the ECM table count and covers essential business functions only, sized for SMB engagements, pilots, and dev/test environments.

ECM (Expanded Coverage Model) is full enterprise breadth, including supporting Corporate domains, sized for Fortune 100 deployments. Attribute depth is identical across both scopes. MVM is not a skeleton, just a smaller surface area.

Cataloging styles: one logical model, three physical layouts



Each model is published as one logical artifact (`model.json`) that deploys to Unity Catalog in any of three physical layouts without any rebuild.

One Catalog puts the entire business in a single catalog. Domains become schemas. Best when one governance boundary fits the organization.

Catalog per Division gives each division (Operations, Business, Corporate) its own catalog. Domains become schemas inside each. Best when divisions are governed separately, with their own admins, retention policies, and classification regimes.

Catalog per Domain gives each domain its own catalog. Subdomains become schemas. Best for federated, data-mesh organizations where each domain owns its own data and governance.

All three styles preserve the same logical relationships, classifications, and metric views. Switching styles at redeploy time does not modify `model.json`.

The forty industries

- **Financial Services**
 - Retail & Investment Banking
 - Payments & Fintech
 - Health Insurance
 - Life & Annuity Insurance
- **Healthcare & Life Sciences**
 - Healthcare Providers
 - Pharmaceuticals
 - Genomics & Biotech
 - Clinical Trials
- **Manufacturing**
 - Industrial Manufacturing
 - Chemical Manufacturing

- Semiconductor
- Automotive
- Construction
- **Retail & Consumer Goods**
 - Retail
 - Grocery
 - E-Commerce
 - Consumer Goods
 - Apparel & Fashion
 - Food & Beverage
 - Restaurants
- **Energy & Utilities**
 - Oil & Gas
 - Energy & Utilities
 - Mining
 - Water & Wastewater
- **Communications, Media & Entertainment**
 - Telecommunications
 - Media & Broadcasting
 - Sports & Entertainment
 - Gaming & Interactive
 - Advertising & Marketing
- **Travel, Transport & Logistics**
 - Airlines
 - Travel & Hospitality
 - Transport & Shipping
 - Shipping & Ports
- **Public Sector, Education & Non-Profit**
 - Education
 - Nonprofit Organizations
 - Legal
 - Waste Management
- **Real Estate & Professional Services**
 - Real Estate
 - Staffing & HR
- **Agriculture**
 - Agriculture

A worked example: the Airlines model

The Airlines ECM is a tier-2 model covering the operational core (flight ops, network and route, fleet, crew, maintenance, ground ops, cargo), the revenue-generating business (sales, distribution, loyalty, customer, payments), and the supporting corporate functions.

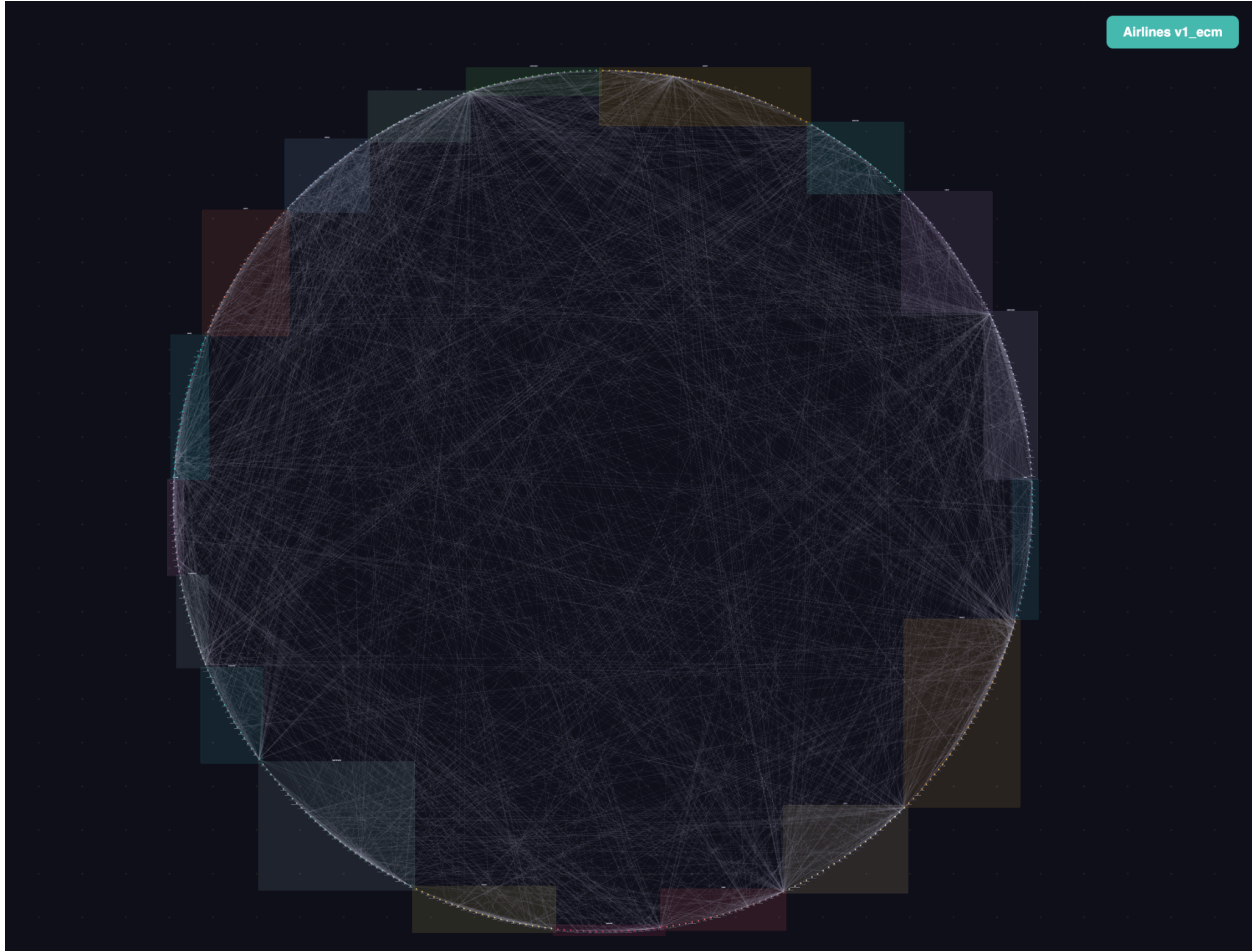
We generated an ECM data model for Airlines that covers all business divisions. Below are the model metrics:

Model Metrics

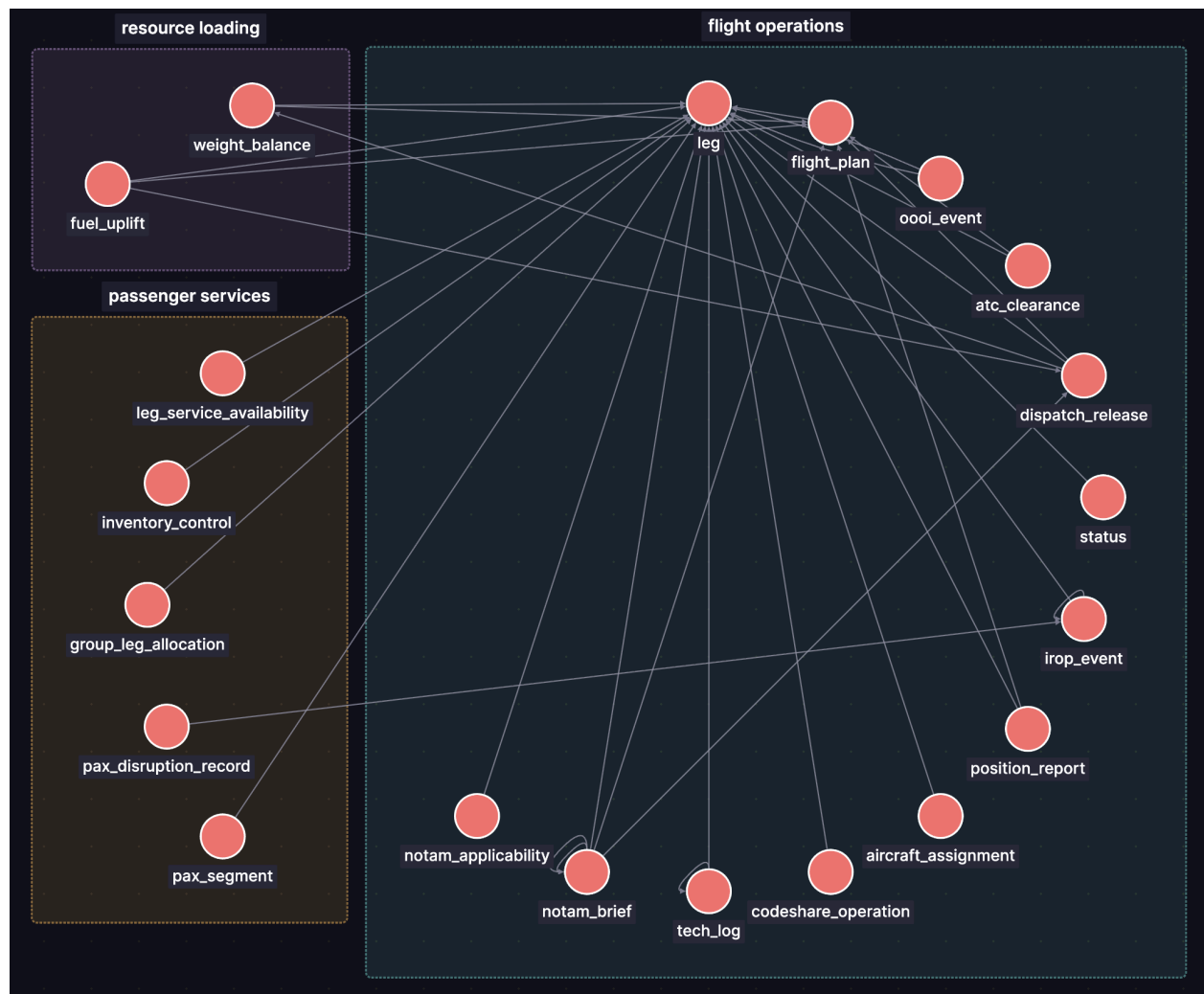
Metric	Value
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Model Scope	ECM (Expanded Coverage Model)
Total Domains	19
Total Subdomains	60
Total Products	420
Total Attributes	17278
Primary Keys	420
Foreign Keys	2877
Avg Attributes/Product	41.1
Metric Views	203

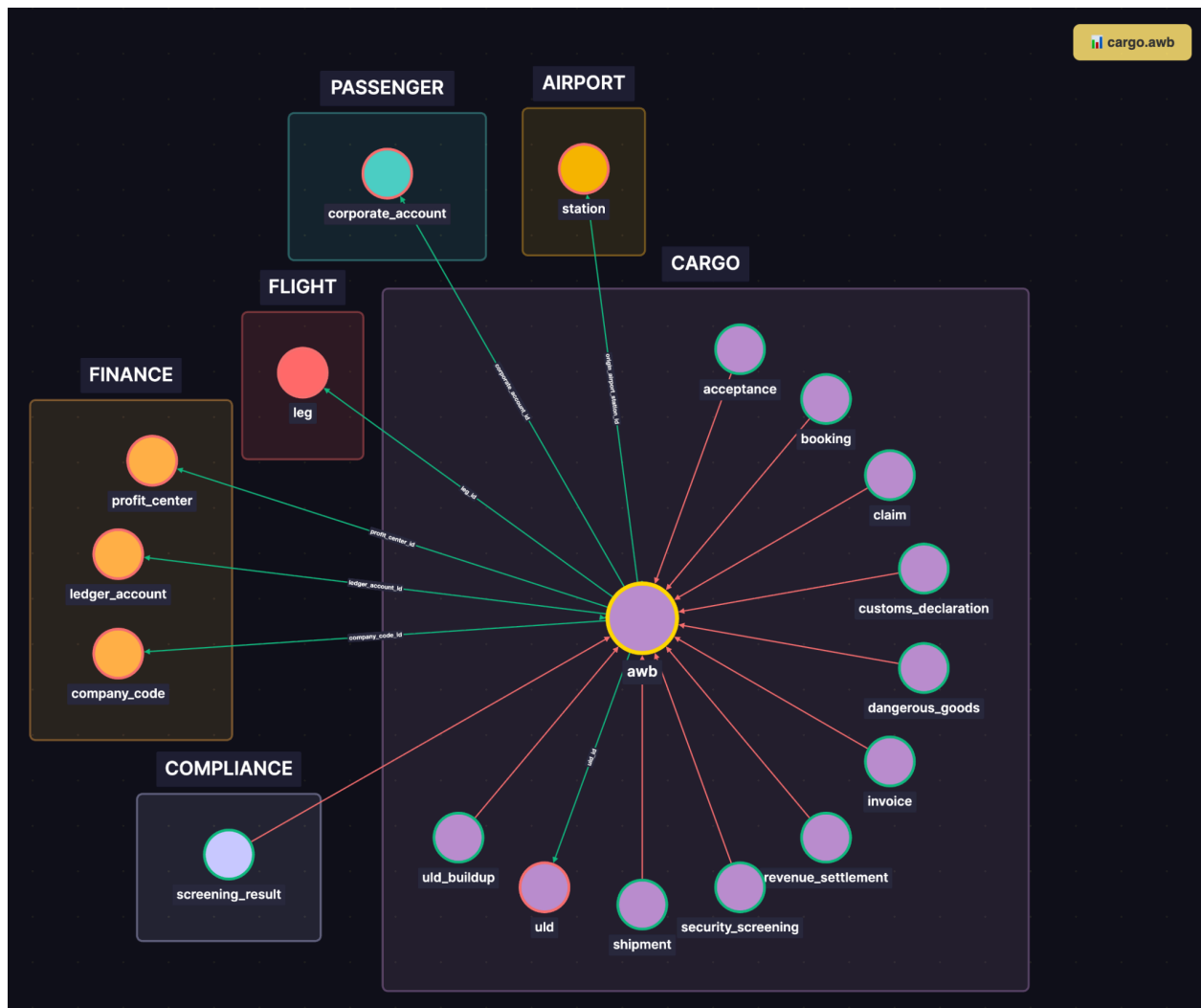
When you visualize this model, it will look like a fully connected DAG (Directed Acyclic Graph) like below:



And this is how the flight domain looks:



When you drill down to, say, the Airway bill data product (awb), it looks like below:



The MVM scope of the same model carries roughly 30 to 50 percent of these table counts at the same attribute depth.

Get started

The library lives at github.com/databricks/vibe-modelling-agent. Pick a model, pick a scope (MVM or ECM), pick a cataloging style, pick a deployment catalog, run the install notebook. A typical MVM install completes in tens of minutes against Databricks Serverless; ECM in under two hours. Both produce a fully deployed Silver layer in Unity Catalog with metric views ready for AI/BI dashboards and AI/BI Genie.

Every model is a starting point. Customers who need to adapt a model to their organization (rename domains, merge or split products, add a missing domain, change naming conventions) can do so using the [Vibe Data Modeling agent](#), which iterates the model through natural-language directives.

Related resources

The announcement blog *Reimagining Data Modeling on the Lakehouse: Introducing Vibe Data Modeling* covers the business case in short form.

The *Vibe Data Modeling* whitepaper (available as an ebook from the Databricks Resource Library) is the full technical treatment for customers who want to understand how the library is built and adapted.

The reference repository at github.com/databricks/vibe-modelling-agent hosts every model, the install notebooks, and the documentation for each industry.

Communications

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Energy

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Financial Services

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HLS

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Manufacturing

Manufacturing

Public Sector

Public Sector

Retail Consumer Goods

Retail Consumer Goods